

# A Split Dedekind Cut Refutes Extension to Every Proper Median Compactification

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**Status.** Candidate counterexample, likely valid, pending expert review. Question 7.2 of Megrelishvili has a negative answer even for a countable rank-one median algebra, the discrete group  $\mathbb{Z}$ , and a compact metrizable linearly ordered median compactification.

## 1 Source question and answer

Megrelishvili proves that every continuous action by median automorphisms extends to the minimal median compactification and then asks whether it extends to *every* proper median compactification [1, Question 7.2]. In the source's notation, the question is: if  $X \in (T_2^m)$  carries its intrinsic median topology and  $G \curvearrowright X$  is jointly continuous by median automorphisms, must that action extend continuously to every proper median compactification of  $X$ ?

*Question 7.2.* Let  $X \in (T_2^m)$  carry its intrinsic median topology, and let  $G \curvearrowright X$  be a jointly continuous action by median automorphisms. Does the action extend continuously to every proper median compactification of  $X$ ? This is true in the linearly ordered case; see [18].

Figure 1: The source question on page 25 of arXiv:2605.16096.

The answer is no. The obstruction is already visible in the elementary theory of split linear orders: a compactification can remember two sides of one nonprincipal cut but not the two sides of its translate.

### Proof intuition

The minimal completion of  $\mathbb{Q}$  fills an irrational Dedekind cut by one point. A different proper median compactification may split that same cut into its left and right sides. Translation by one carries both sides toward the single unsplit cut one unit later. Continuity would therefore identify the two split points, whereas a group element must act by a homeomorphism. The example satisfies every hypothesis because splitting a cut outside  $\mathbb{Q}$  neither changes the topology induced on  $\mathbb{Q}$  nor disturbs its order median.

**Theorem 1.** *There exist a median algebra  $X \in (T_2^m)$  carrying its intrinsic median topology, a jointly continuous action of a topological group  $G$  by median automorphisms, and a compact metrizable proper median compactification  $X \hookrightarrow K$  to which the action does not extend. One may take*

$$X = \mathbb{Q}, \quad G = \mathbb{Z}, \quad n \cdot q = q + n.$$

*In particular, Question 7.2 is false even for linearly ordered median algebras.*

## 2 The split compactification

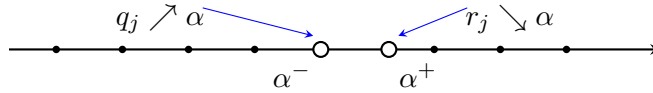
Let  $\overline{\mathbb{R}} = [-\infty, +\infty]$  with its usual compact order and fix  $\alpha = \sqrt{2}$ . Define

$$K = (\overline{\mathbb{R}} \setminus \{\alpha\}) \cup \{\alpha^-, \alpha^+\}$$

by leaving the order unchanged away from  $\alpha$  and declaring

$$x < \alpha^- < \alpha^+ < y \quad \text{whenever} \quad x < \alpha < y.$$

Thus  $\alpha^-$  and  $\alpha^+$  are adjacent. Equip  $K$  with its order topology and order median.



**Lemma 1.**  *$K$  is a compact metrizable topological median algebra, and the natural inclusion  $\mathbb{Q} \hookrightarrow K$  is a dense topological median embedding. Consequently it is a proper median compactification of  $(\mathbb{Q}, \tau_m)$ .*

*Proof.* The order on  $K$  has first and last elements and is order complete. Indeed, collapse the split pair to  $\alpha$ . Suprema are inherited from  $\overline{\mathbb{R}}$ , with the only choice being  $\alpha^-$  or  $\alpha^+$  when the projected supremum is  $\alpha$ ; the membership of  $\alpha^+$  resolves that choice. A complete linear order with endpoints is compact in its order topology. Splitting only one point preserves second countability, so  $K$  is metrizable.

Every nonempty open interval of  $K$  contains a rational. In particular, rationals below  $\alpha$  approach  $\alpha^-$  and rationals above  $\alpha$  approach  $\alpha^+$ ; hence  $\mathbb{Q}$  is dense. Since  $\alpha \notin \mathbb{Q}$ , intersecting an open order interval of  $K$  with  $\mathbb{Q}$  produces an ordinary open interval or ray in  $\mathbb{Q}$  (possibly cut at the irrational number  $\alpha$ ). Conversely, ordinary rational intervals are such intersections. Thus the subspace topology is exactly the usual order topology on  $\mathbb{Q}$ .

The inclusion is order preserving, hence preserves the middle-of-three median. Finally, every LOTS is a topological median algebra: min and max are continuous, as is immediate from inverse images of open rays, and

$$\text{med}(x, y, z) = \max\{\min(x, y), \min(y, z), \min(z, x)\}.$$

For a chain, the intrinsic median topology equals the interval topology [1, Proposition 4.4(1)]. Therefore  $\mathbb{Q} \in (T_2^m)$  and the embedding is a proper median compactification in the exact sense of the source.  $\square$

## 3 Translation collapses the split pair

*Proof of Theorem 1.* Give  $G = \mathbb{Z}$  the discrete topology and let it act on  $X = \mathbb{Q}$  by translations. Every translation is an increasing homeomorphism and therefore a median automorphism. Since  $G$  is discrete, the action  $G \times \mathbb{Q} \rightarrow \mathbb{Q}$  is jointly continuous.

Choose rational sequences

$$q_j \nearrow \alpha, \quad r_j \searrow \alpha.$$

In  $K$  they satisfy

$$q_j \longrightarrow \alpha^-, \quad r_j \longrightarrow \alpha^+.$$

Suppose the action extended continuously to  $K$ , and denote the extension of translation by one by  $T$ . The cut  $\alpha + 1$  is not split in  $K$ , so

$$q_j + 1 \longrightarrow \alpha + 1, \quad r_j + 1 \longrightarrow \alpha + 1.$$

Continuity gives

$$T(\alpha^-) = \alpha + 1 = T(\alpha^+).$$

Thus  $T$  is not injective. On the other hand, in any  $\mathbb{Z}$ -action the maps corresponding to 1 and  $-1$  are mutual inverses, so  $T$  must be a homeomorphism. This contradiction proves that no extended action exists.  $\square$

The MMC of  $\mathbb{Q}$  is the ordinary Dedekind compactification  $\overline{\mathbb{R}}$ , on which translations do extend. The factor map  $K \rightarrow \overline{\mathbb{R}}$  simply collapses  $\alpha^-$ ,  $\alpha^+$ , illustrating exactly why minimality is compatible with the counterexample.

## 4 Exact invariance criterion

The same argument gives a useful classification. Let  $D$  be the Dedekind compactification of a dense chain  $X$ , let  $S$  be a set of two-sided nonprincipal cuts, and let  $K_S$  be the compact order obtained by replacing each  $s \in S$  by adjacent points  $s^- < s^+$ . Write  $p : K_S \rightarrow D$  for the collapse map.

**Proposition 1.** *Let  $h$  be an increasing automorphism of  $X$ , and let  $\hat{h}$  be its canonical extension to  $D$ . Then  $h$  extends to an order homeomorphism of  $K_S$  if and only if*

$$\hat{h}(S) = S.$$

*Consequently, an increasing group action on  $X$  extends to  $K_S$  exactly when  $S$  is invariant under the induced action on the Dedekind cuts.*

*Proof.* Suppose  $H : K_S \rightarrow K_S$  is a homeomorphism extending  $h$ . The continuous maps  $pH$  and  $\hat{h}p$  agree on the dense copy of  $X$ , hence agree everywhere. The fibers of  $p$  have size two exactly over  $S$ . Since  $H$  is bijective and intertwines the collapse maps, it preserves the set of two-point fibers; therefore  $\hat{h}(S) = S$ .

Conversely, if  $\hat{h}(S) = S$ , define  $H$  on unsplit points by  $\hat{h}$  and set

$$H(s^-) = \hat{h}(s)^-, \quad H(s^+) = \hat{h}(s)^+.$$

This is an order isomorphism of compact LOTS, hence a homeomorphism, and it extends  $h$ .  $\square$

For Theorem 1,  $S = \{\sqrt{2}\}$  is not invariant under  $q \mapsto q+1$ . Proposition 1 shows that nonextension is the exact orbit obstruction, rather than a sequential accident.

## 5 What the cited ordered theorem actually gives

The sentence following Question 7.2 says that the desired conclusion is true for linearly ordered spaces and cites [2]. The relevant result in that paper assumes that the chosen proper ordered compactification already admits extended continuous  $g$ -translations; it then proves that the resulting action is jointly continuous. The underlying ordered theorem is stated even more explicitly in [3, Theorem 3.18]: the ordered proximity must be  $G$ -invariant, or equivalently the compactification must already be a  $G_{\text{discr}}$ -compactification.

Thus the cited theorem is a *separate-to-joint continuity* result, not an extension-existence result for every compactification. Proposition 1 identifies precisely the missing hypothesis: the split set must be invariant. With that hypothesis restored, the ordered theorem is fully consistent with this counterexample.

## 6 Verification and review focus

The proof is noncomputational. It reduces to four elementary checks made explicit above:  $K$  is a compact complete order; the dense embedding induces the intrinsic order topology on  $\mathbb{Q}$ ; the middle-of-three operation is continuous; and the two one-sided rational sequences have distinct limits before translation but the same limit afterward. Human review should also compare the literal sentence after Question 7.2 with the  $G$ -invariance hypothesis in [3, Theorem 3.18].

## 7 Scope and novelty

The universal question is settled negatively by a rank-one example. The compactification and acting group are both metrizable/countable in the strongest elementary sense, so no higher-rank or nonmetrizable pathology is involved.

Local run indexes were searched by arXiv id, exact title and question, and split-cut/translation terminology. Bounded primary-source searches found the source papers and standard ordered-compactification criteria but no explicit answer to Question 7.2 or this construction. Since split orders are classical, novelty is asserted only relative to the specific 2026 question and remains subject to expert review.

## References

- [1] M. Megrelishvili, *Intrinsic uniform structure on median algebras*, arXiv:2605.16096 (2026), arXiv:2605.16096.
- [2] M. Megrelishvili, *Circular orders: Topology and continuous actions*, arXiv:2512.17314 (2025), arXiv:2512.17314.
- [3] M. Megrelishvili, *Maximal equivariant compactifications*, Topology Appl. **329** (2023), 108372; arXiv:2201.13426.